



What is self-rated health and why does it predict mortality? Towards a unified conceptual model[☆]

Marja Jylhä

School of Public Health, University of Tampere, FIN-33014, Finland

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ABSTRACT

The association of self-rated health with mortality is well established but poorly understood. This paper provides new insights into self-rated health that help integrate information from different disciplines, both social and biological, into one unified conceptual framework. It proposes, first, a model describing the health assessment process to show how self-rated health can reflect the states of the human body and mind. Here, an analytic distinction is made between the different types of information on which people base their health assessments and the contextual frameworks in which this information is evaluated and summarized. The model helps us understand why self-ratings of health may be modified by age or culture, but still be a valid measure of health status. Second, based on the proposed model, the paper examines the association of self-rated health with mortality. The key question is, what do people know and how do they know what they know that makes self-rated health such an inclusive and universal predictor of the most absolute biological event, death. The focus is on the social and biological pathways that mediate information from the human organism to individual consciousness, thus incorporating that information into self-ratings of health. A unique source of information is provided by the bodily sensations that are directly available only to the individual him- or herself. According to recent findings in human biology, these sensations may reflect important physiological dysregulations, such as inflammatory processes. Third, the paper discusses the advantages and limitations of self-rated health as a measure of health in research and clinical practice. Future research should investigate both the logics that govern people's reasoning about their health and the physiological processes that underlie bodily feelings and sensations. Self-rated health lies at the cross-roads of culture and biology, therefore a collaborative effort between different disciplines can only improve our understanding of this key measure of health status.

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Introduction

Hardly any other measure of health is more widely used and more poorly understood than self-rated health (or self-assessed health, or self-perceived health). It is based on asking individuals to evaluate their health status on a four- or five-point scale, or to compare their health status with that of age peers. This simple question has been one of the most frequently employed health indicators in sociological health research since the 1950s (Garrity, Somes, & Marx, 1978; Maddox, 1962; Suchman, Phillips, & Streib, 1958). In epidemiological and medical research it began to gain currency when Singer, Garfinkel, Cohen, and Srole (1976) and then

Mossey and Shapiro (1982) and Kaplan and Camacho (1983) demonstrated its association with mortality. It is frequently used by economists as well (Crossley & Kennedy, 2002; Juerges, 2007). Data on self-rated health are collected in major national and international surveys, such as the World Value Survey and European Value Survey (Inglehart et al., 2003; Mansyur, Amick, Harrist, & Franzini, 2008), the National Health and Nutrition Examination Survey in the US and the SHARE in Europe. It is included in the SF-36 survey instrument (Ware & Gandek, 1998) and recommended as a standard part of health surveys (Robine, Jagger, & the EURO-REVES Group, 2003; WHO, 1996). It has also been recommended as a tool for disease risk screening (May, Lawlor, Brindle, Patel, & Ebrahim, 2006) and clinical trials (Fayers & Sprangers, 2002).

The exact wordings and response options of self-rated health questions vary. In the question that is most widely used in the US, responses are set out on a scale from excellent through very good, good and fair to poor, while the options recommended by WHO (1996) and the EURO-REVES 2 group (Robine et al., 2003) are very

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E-mail address: marja.jylha@uta.fi

good, good, fair, bad and very bad; there is also a version that uses the options very good, fairly good, average, fairly bad, and bad (Heikkinen, Waters, & Brzezinski, 1983; Jylhä, Guralnik, Ferrucci, Jokela, & Heikkinen, 1998). Although the levels and distributions are not directly comparable between the measures, they represent parallel assessments of the same phenomenon, and show basically concordant answers (Jürges, Avendano, & Mackenbach, 2007; Jürges, Avendano, & Mackenbach, 2008). In addition to these global questions, a comparative measure is sometimes used (“How would you assess your health status in comparison with other people of your age that you know? Is it better, the same or worse?”). The main focus in this paper is on global self-rated health, although most of what will be said also applies to the comparative version. Where appropriate, the differences in the behavior of these versions are pointed out.

In 1958, Edward A. Suchman, Bernard S. Phillips and Gordon F. Streib concluded that “self-ratings of health measure something different than physician’s ratings – what we have called ‘perceived’ or ‘subjective’ health as opposed to ‘actual’ or ‘objective’ health – but that depending upon one’s hypothesis such a self-rating may or may not be valid”. Since then, the relationship of self-rated health to different individual and environmental characteristics has been explored in several hundred studies, and its associations with mortality have been the focus of concern in around one hundred. Several researchers, most notably Ellen Idler (Idler, 1979; Idler & Benyamini, 1997; Idler & Kasl, 1991; Idler, Leventhal, McLaughlin, & Leventhal, 2004) but also others (Benyamini, Leventhal, & Leventhal, 1999; Ferraro, 1980; Ferraro & Kelley-Moore, 2001; Knäuper & Turner, 2003; Tissue, 1972) have shed light on the unique characteristics of the self-rated health measure and its role in predicting mortality. Nevertheless, studies repeatedly highlight our deficient and fragmentary understanding of the nature of this indicator: it is just not clear what exactly self-rated health measures and why it has such a strong and constant association with mortality.

The motivation for this paper comes from the conviction, based on many years of work around this theme (Jylhä, 1994; Jylhä et al., 1998; Jylhä, Leskinen, Alanen, Leskinen, & Heikkinen, 1986; Jylhä, Volpato, & Guralnik, 2006; Leinonen, Heikkinen, & Jylhä, 2001), that our difficulties in understanding self-rated health stem not so much from a lack of empirical information as from conceptual and theoretical vagueness and poor integration of knowledge. Self-rated health, an individual and subjective conception that is related to the strongest biological indicator, death, constitutes a cross-road between the social world and psychological experiences on the one hand, and the biological world, on the other. Most empirical studies on self-rated health come from the epidemiological tradition that focuses on the statistical associations of variables rather than on the processes from which the variables emerge. There is an obvious need for more comprehensive approaches.

The purpose of this paper is to bring together the cognitive perspective and the epidemiological perspective to self-rated health into a common conceptual framework. The proposed model helps us understand the individual process of health assessment, its biological basis and its social and cultural contexts and, ultimately, the association of self-rated health with mortality. It can also help researchers from different backgrounds interpret their empirical findings and generate new research.

The paper has three parts. First, I discuss the notion of self-rated health and suggest a model that describes the process of evaluation in which self-ratings are produced. The starting point is to acknowledge that self-rated health originates in an active cognitive process and cannot be properly understood without understanding this process. The purpose is to clarify the nature of the different types of information that people have about their health and the way in which they use this information in their self-ratings. Second,

based on this model, I proceed to examine the relationship of self-rated health with death. Here, new research at the interface of cognitive neuroscience, neurobiology and immunology is discussed to develop hypotheses on the pathways that mediate information on bodily status to individual consciousness. Third, I discuss some of the advantages and limitations of self-rated health as a measure of health in research and clinical practice. Finally, I suggest some directions for future research. Throughout, the new thoughts and hypotheses are developed and validated against earlier research findings.

The process of health assessment

Self-rated health differs from most indicators of health in that its origins lie in an active cognitive process that is not guided by formal, agreed rules or definitions. It can be understood as “...a summary statement about the way in which numerous aspects of health, both subjective and objective, are combined within the perceptual framework of the individual respondent” (Tissue, 1972, p. 93). What, then, are the relevant aspects, how are they combined, and in what sort of framework?

By referring to self-assessment of health as a cognitive process, I do not mean to suggest that these self-ratings are always based on logical steps of conscious reasoning. There are good reasons to believe that the answers to the self-rated health question sometimes are more intuitive, sometimes more consciously reflective. However, the evaluation necessarily involves the processing of information, interpretation of meanings, and selection. Although an individual, subjective process, it is embedded in a given social and cultural environment and makes use of the conceptual resources and patterns of representations provided by this environment. The evaluation process is outlined in Fig. 1.

Several earlier studies have suggested models for self-rated health. Most of them aim at developing statistical explanatory analyses (e.g. Fylkenes & Forde, 1992; Moum, 1992; Tornstam, 1975). The model by Knäuper and Turner (2003), however, describes the cognitive “flow” from the question to the individual self-rating. They start out from the premise that the individual’s judgement is based on a mental representation of personal health that consists of semantic and episodic knowledge, as well as information about change. The model also includes the socio-demographic and personality factors that influence the outcome of the evaluation. The main focus of their paper is on the design of questionnaires, and it does not include a detailed discussion of different parts of the evaluation process. Some of the similarities and differences between their model and the one presented here will be discussed later in this paper.

For analytic purposes, the model proposed here (Fig. 1) distinguishes different stages in the process of self-assessment; these are described in the right column of Fig. 1. First, the person has to recognize the meaning of “health” and identify the components that should be taken into account as components of “my health status”; second, he or she has to consider the way in which those components should be taken into account; and finally, one has to decide which of the levels in the preset scale summarizes them best. At each stage, the evaluation is influenced by contextual factors, described in the left column of Fig. 1. The final outcome of the evaluation, then, is dependent on the individual’s understanding of what “health” is and the review of the components of “my health”, but also on the contextual frameworks within which the components of health are considered, and the ways in which the preset evaluation scale is used. The process of evaluation is discussed in more detail below.

In the first stage of health evaluation, when the individual is asked to give a general rating of his or her health, an apparent

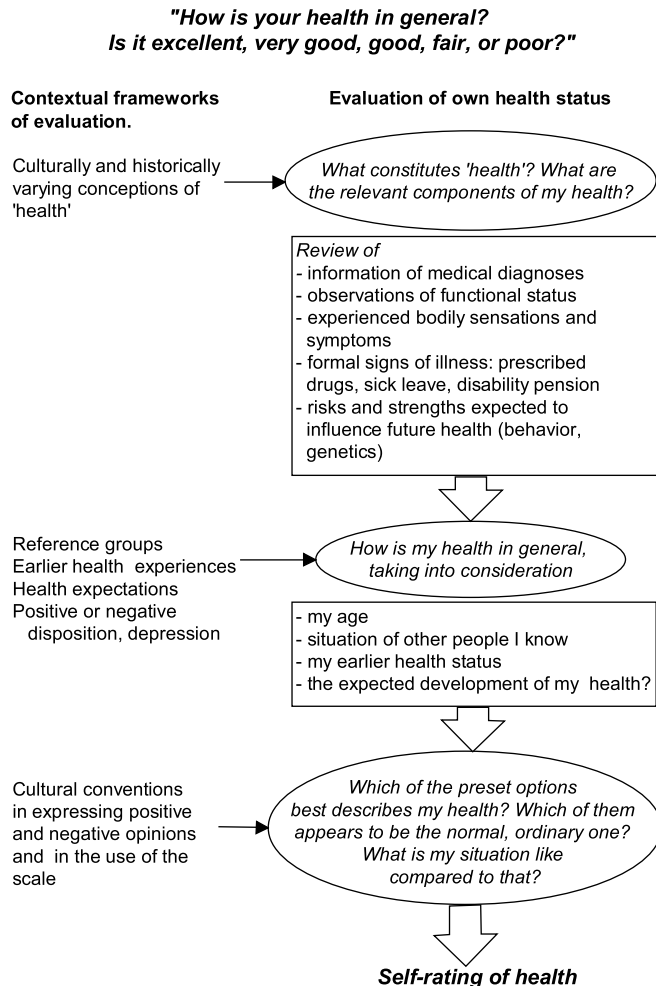


Fig. 1. The process of individual health evaluation.

problem is that there is no single universally agreed definition or direct measure of “health” or “health status”. In fact, “health” belongs to the realm of everyday talk rather than any exact scientific vocabulary. Both in research and in clinical practice, “health” is used as a generic term that is empirically approached by various indicators such as medical diagnoses, functional status, experienced symptoms, and laboratory values; there is no standard rule or equation for how to develop these into an exhaustive global indicator of “health”. People have considerable freedom to decide what information to base their evaluations on. Still, it can be assumed that people do not select this information arbitrarily, but their choices are guided by the general cultural understanding of the phenomena or categories that belong to the realm of “health”. This understanding is obviously historically and culturally variable. In today’s industrialized societies, lay people and the medical profession probably share a fairly similar general understanding of what constitutes “health” and what information is essential to describing it – although the interpretations and conclusions drawn from this information may differ.

Components of “health” in self-assessments

The model suggested here assumes that the direct components of self-assessed health consist of individually relevant information about conditions, states, sensations etc. that the respondent understands as descriptors of “health”. From this perspective, the

role of “non-health” individual and social characteristics such as social class, education, standard of living, social networks, social capital, and the quality of the neighborhood (Kawachi, Kennedy, & Glass, 1999; Krause, 1996; Mansyur et al., 2008; Schultz, O’Brien, & Tadesse, 2008; Singh-Manoux et al., 2006) is different. They are likely to show a statistical association with self-rated health not because they directly describe health status but, first, because they influence the likelihood of different objective health conditions being used as a basis for self-ratings (Manderbacka, Lundberg, & Martikainen, 1999); and, second, because they shape the frameworks of evaluations.

Several studies have shown that objective clinical or physiological states, consequential as they may be for the individual’s future, can only be taken into account in self-assessments and can only form a basis for these assessments if they are known to the individual, directly or indirectly, as diagnostic labels or as signs or symptoms interpreted as health-related (Campbell et al., 2008; Idler & Kasl, 1995; Idler et al., 2004). The information that is available to the individual about his or her health varies by nature and by source. Knowledge about medical diagnoses usually comes from a physician or other health care professionals. Indirect formal information about medical conditions also includes prescribed medical drugs and other treatments, sick leaves or disability pensions. Functional status and disability can be observed by the individuals themselves and by other people around them. Various symptoms and sensations, such as pains, aches, fatigue, dizziness, and low spirits, again, are signals from the individual body and mind that are directly available only to the person him- or herself, and only indirectly, through language or behavior, to others. The importance of subjective information was emphasized also by Knäuper and Turner (2003). Yet another type of information about health problems includes conceptions about the time-line, consequences and controllability of different conditions (Leventhal, Idler, & Leventhal, 1999).

Recent studies imply that in the current ethos of individual responsibility for healthy lifestyles, younger cohorts increasingly understand health behaviors not only as (positive or negative) risk factors, but as direct components of health status and that they take them into account as such in their health evaluations (Chen, Cohen, & Kasen, 2007; Manderbacka et al., 1999). In the future, new categories of health information may emerge to influence self-ratings of health. An example is provided by the changing role of genetic information. In twin studies, the genetic component has been found to explain some 60% of the variation seen in self-rated health (Leinonen et al., 2005; Silventoinen, Posthuma, Lahelma, Rose, & Kaprio, 2007). Statistical analyses show that this effect is not direct, but mediated through different conditions on which information is available to the individual, such as disease severity, mobility and symptoms (Leinonen et al., 2005). In the future, as people learn more about their genetic constitution, it is possible that genetic risk factors will be taken into account in self-ratings as important components of health.

The framework of evaluation

The next step in the process of health assessment, analytically separate although not necessarily distinct in individual experience, includes the consideration of how to take the specific relevant aspects of health status into account and how to summarize them into “my health in general”. In a few studies respondents have been asked, once they have provided their health rating on a given scale, to explain why they chose this particular answer, or what went through their mind when they said what they did (Borawski, Kinney, & Kahana, 1996; van Dalen, Williams, & Gudex, 1994; Idler, Hudson, & Leventhal, 1999; Krause & Jay, 1994; Simon, De Boer, Bosma, & Mackenbach, 2005), or to rate the importance of different

factors to their health (Benyamini, Leventhal, & Leventhal, 2003). These studies do not support the suggestion by Knäuper and Turner (2003) that in their evaluations people deliberately focus on mortality-relevant information and disregard mortality-irrelevant information. Instead, in line with the findings of quantitative studies, they indicate that different aspects of objective health and functioning, both severe and less severe, constitute the core set of factors underlying self-rated health, but they also demonstrate the diverse, selective ways in which these factors are used. It is evident that for their own self-assessments, people do not consider different health factors in general, abstract terms, but in the context of their own situation, own health status and health history. Qualitative analyses of interview episodes dealing with self-rated health show that different aspects of health may seem incommensurable and even contradictory to the individual; “in this respect” they would arrive at one conclusion, “in that respect” at another. Often it is only through a complex process of reasoning and negotiation that this genuinely multidimensional phenomenon of “my health” can be fitted into the preset scale (Jylhä, 1994).

It may well be that in the respondents' minds, the scale from very good to poor health does not represent a linear continuum. In his analysis of the “goodness” of bodily health, the philosopher G.H. von Wright regards “good” health as the state of normalcy, a state where nothing is wrong. Good health is the baseline that does not normally need to have a cause; it is the deviation from this normal state that must have some cause and that requires explanation (von Wright, 1963). Empirical analyses indicate that “good” and less than good self-rated health differ more fundamentally than simply in terms of the number of specific underlying health problems. The reasons that lead one person to rate his or her health as poor do not necessarily represent a mirrored picture of the reasons why another person rates his or her health as good (Benyamini, Blumstein, Lusky, & Modan, 2003; Kaplan & Baron-Epel, 2003; Krause & Jay, 1994; Shooshtari, Menec, & Tate, 2007; Simon et al., 2005; Smith, Shelley, & Dennerstein, 1994).

Previous and present health experiences are likely to influence both the range of factors that a person reviews as potential components of self-rated health and the way in which they are taken into account (Heller, Ahern, Pringle, & Brown, 2008). Someone who has a serious disease, for example breast cancer, will probably consider it in the process of assessing their health, while someone who has fully recovered from breast cancer may see this as a positive component of their health. However, a person who has never experienced breast cancer is unlikely to give active consideration to this disease in her self-rating. On the other hand, a recent study demonstrates the role of adaptation in health assessments: in response to new chronic conditions, older individuals with greater pre-existing comorbidity were less likely to further decrease their health assessments than younger, healthier individuals (Heller et al., 2008).

One obvious yardstick for health evaluations is provided by comparisons to a chosen reference group (Festinger, 1954; Merton, 1957), either to known individuals or to a general image or stereotype of a certain group (Suls, Marco, & Tobin, 1991). Here, age constitutes an important frame of evaluation, particularly because the frequency of health problems differs between age groups. This is also true when the non-comparative, global question of self-rated health is used. Young people, who assess their health against the age-specific expectation of a good health status, may interpret the information of their own health differently from older people who expect to have some degree of health problems. It can be assumed that for younger people, being free of serious disease constitutes the baseline of health assessment. In this situation, factors such as favorable health behavior and physical fitness serve as arguments for a positive evaluation, and chronic disease or functional

limitations for a negative assessment. In old age the context of evaluation is more complex. For most old people it requires negotiation between the normative non-problematic category of “good” health and experienced problems in health and functioning. It is no wonder that old people more often than younger people base their self-assessments of health on social and temporal comparisons to earlier health or future expectations, or, above all, on comparisons with age peers (Cheng, Fung, & Chan, 2007; Fayers, Langston, Robertson, & on behalf of the PRISM Trial, 2007; Idler et al., 2004; Suls et al., 1991; Tissue, 1972). These comparisons with specific people, but also with deceased members of one's own birth cohort, or with negative stereotypes of old age, are likely to lead to a lower aspiration level concerning one's own “good” health, allowing for positive assessments in spite of health and functional problems (Tornstam, 1975).

The importance of age as an evaluation framework is demonstrated by its seemingly paradoxical behavior in population studies. In cross-sectional age group comparisons, global self-rated health does not decrease with advancing age to the same extent as chronic conditions and disability increase (Jylhä, Guralnik, Balfour, & Fried, 2001), and in follow-up studies with the same respondents it may remain the same even if the direct question about change in health status indicates a worsening state (Leinonen, Heikkinen, & Jylhä, 1998). For a given level of measured health conditions, older people usually assess their health more positively than younger people (Ferraro, 1980). Controlling for other health indicators in multivariate analyses usually leads to a negative correlation between age and poor health (Jylhä et al., 2001; Mulsant, Ganguli, & Seaberg, 1997). When asked to compare their health with that of age peers, old people give more positive ratings than the middle-aged (Denning et al., 1998; Vuorisalmi, Lintonen, & Jylhä, 2006).

Standards of “good” health are also likely to change over time. Two studies recently showed decreasing time trends in self-rated health in populations with clearly improving life expectancies (Jagger et al., 2007; Zack, Moriarty, Stroup, Ford, & Mokdad, 2004). Rather than declining trends in physical health, these findings probably reflect a secular change towards a higher aspiration level, or higher expectations concerning one's health status.

Different cultures provide different frameworks for health evaluations. Italians assessed their health more positively than Finns (Jylhä et al., 1998) but more negatively than French people (Desquelles, Egidi, & Salvatore, 2009), Dutch men more positively than Lithuanian men (Appels, Bosma, Grabauskas, Gostautas, & Sturmans, 1996), non-Hispanic whites in Colorado more positively than Hispanics (Shetterly, Baxter, Mason, & Hamman, 1996), and Arabs in Israel more positively than Jews (Baron-Epel et al., 2005). These differences remained after controlling for other available social and health characteristics, although the associations between self-rated health and these characteristics were largely similar between the respective countries. These differences may reflect, first, real differences in objective health status that are not covered by the specific measures available in the analyses but are taken into account in self-ratings; second, differences in levels of knowledge about one's own health that underlie self-rated health; or, third, differences in ways of making one's evaluations. The finding that it is not always the lower mortality and higher social status group that reports more positive self-rated health suggests that genuine differences in the process of evaluation play at least some role here.

Cultural patterns of experiencing and reporting bodily symptoms are well-known in health sociology (Zborowski, 1952; Zola, 1966), and the association of such symptoms as pain with self-rated health varies between cultures (Gureje, von Korff, Simon, & Gater, 1998). These differences may have several origins. First, the “baseline morbidity” in a population group may influence the understanding of which symptoms or conditions warrant attention,

and how they are attributed (Kaplan & Baron-Epel, 2003). Second, the response styles vary. People living in different cultures differ in their willingness to present clearly positive or negative pictures of themselves (Baron-Epel et al., 2005; Bzostek, Goldman, & Pebley, 2007). Also, it seems that there are cultural differences in ways of using preset response scales, for instance in the choice of options at either extreme of the scale. Linguistic factors, particularly the different connotations of the preset response options, play a role here: in standard translations to Spanish and Russian, the middle options “fair” or “average” seem to describe normal health and replace “good” as an anchor point for evaluations (Bzostek et al., 2007; Palosuo, Uutela, Zhuravleva, & Lakomova, 1998).

The often implicit contextual frameworks that shape the ways in which people evaluate their own health are not limited to those discussed above, but they are the most important ones in empirical research. These frameworks are difficult to measure directly in quantitative studies; usually they are indirectly represented by indicators such as age, nationality or ethnic group. The framework of evaluation may also be shaped by individual characteristics such as an optimistic or pessimistic general disposition (Brissette, Leventhal, & Leventhal, 2003). Depression is an important component of health in self-ratings (Han & Jylhä, 2006; Schnittker, 2005), but it also modifies the framework of evaluation and may lead to a more negative interpretation of one's own situation.

The model proposed here is intended not to describe causal effects on objective physical health status, but the cognitive process in which health status is evaluated in response to an interview question. The distinction between components of health and frameworks that shape the way in which these components are evaluated should not be understood as an absolute or ontological one, but primarily as an analytical tool. The main purpose of the model is to make understandable the nature of self-rated health as a subjective and contextual self-assessment, and as an indicator of objective somatic and mental state, at one and the same time. The following discusses in more detail the implications of this “dual nature” to the association of self-rated health with mortality and to the use of the variable in research and clinical practice.

Why does self-rated health predict mortality?

Studies on the association of self-rated health with mortality have been earlier reviewed by Benyamini and Idler (1999), DeSalvo, Bloser, Reynolds, He, and Muntner (2006) and Idler and Benyamini (1997); these reviews cover the work done before 2003. A Medline search conducted in January 2008 for the period from 2003 to 2007 yielded an additional 32 population studies, although some of them share the same data. The latest findings repeat, confirm and expand on the findings of earlier studies, but the basic message remains unchanged: self-rated health, adjusted for age, shows a graded association with mortality, and this association is usually attenuated but seldom disappears when other factors, primarily health indicators, are controlled for. The association has been confirmed in very different populations: in young people (Larsson, Hemmingson, Allebeck, & Lundberg, 2002) and in nonagenarians (Nybo et al., 2003); in countries such as China (Yu et al., 1998), Denmark (Nielsen et al., 2008), Finland and Italy (Jylhä et al., 1998), Germany (Heidrich, Liese, Lowel, & Keil, 2002), Indonesia (Frankenberg & Jones, 2004) and Japan (Ishizaki, Kai, & Imanaka, 2006); in different patient groups such as older people in emergency department (Wong, Wong, & Caplan, 2007), patients with coronary artery disease (Bosworth et al., 1999) and in people with mild to moderate cognitive decline (Walker, Maxwell, Hogan, & Ebly, 2004). In fact, this association seems to be universal in all the populations studied. However, recent studies have revealed *relative* differences between population groups: self-rated health seems to be a stronger

predictor of mortality in younger than in older age groups, and in the young-old than in old-old (Benyamini, Blumstein, et al., 2003; Franks, Gold, & Fiscella, 2003), in men than in women (Deeg & Kriegsman, 2003; Dowd & Zajacova, 2007; Huismans, van Lenthe, & Mackenbach, 2007; Spiers, Jagger, Clarke, & Arthur, 2003), in higher than in lower socio-economic groups (Dowd & Zajacova, 2007; Huismans et al., 2007), and when ethnic groups are compared in the US, stronger for whites than other groups (Lee et al., 2007).

Self-rated health is said to be “an independent predictor of mortality”. It may be useful to specify what this means. Ultimately, death is a biological event, and its proximal causes have to be found in the failure of vital physiological processes. For self-rated health, being a “predictor of mortality” hardly implies a major role in a biological causative chain leading to death. Rather, self-rated health can be understood as a condensed summary of information about bodily conditions that in one way or another are involved in these biological chains. In other words, the hypothesis is that self-rated health is a statistical (rather than a causative) predictor of mortality because of its ability to reflect the state of the human organism; the accuracy with which it can reflect this state depends on the comprehensiveness and accuracy of the information that the individual incorporates into the self-rating.

Several studies show that the association of self-rated health with mortality is stronger when the cause of death is a condition of which the respondent is likely to have known when giving the health assessment, such as diabetes in older people (Benjamins, Hummer, Eberstein, & Nam, 2004), or alcohol and drug related causes in young men (Larsson et al., 2002), and much weaker for sudden causes such as violence. The importance of knowing is demonstrated in several other empirical studies as well. Idler et al. (2004) showed that among people with circulatory disease, self-rated health was a significant “independent” predictor of mortality only in the subgroup who knew about their diagnosis, but in those who were not aware of the disease the association did not reach significance. Pijls, Feskens, and Kromhout (1993) found that self-rated health was a strong predictor of the fatality of diagnosed chronic diseases, but did not predict the incidence of the same specific chronic diseases. All these studies imply that the more accurate the individual's information about his or her objective bodily condition, the better the prognostic value of self-rated health. Differences in the level and accuracy of information about own health status may partly explain the variation in the explanatory power of self-rated health between social, cultural and age groups. With advancing age, people are more likely to experience new, possibly life-threatening health events during the follow-up, and these conditions are obviously not taken into account in self-ratings at the baseline; this may contribute to a weaker association with mortality in older age groups.

The understanding of self-rated health as a carrier of information about objective bodily condition is supported, indirectly, by findings that in multivariate statistical models, the association of self-rated health with mortality is likely to be the weaker, the better and more comprehensive the other health measures included in these models. However, if not only self-rated health but also the other health variables included in the analysis are based on self-reports, they are likely to be modified by the same evaluation framework, for instance by the same optimistic or pessimistic disposition. Therefore, the rather few studies that have used clinically verified health data and objectively measured indicators are particularly valuable. The five-year mortality analysis in the Cardiovascular Health Study had access to probably the most comprehensive health information ever used in mortality analysis together with self-rated health (Fried et al., 1998). In this study, self-rated health showed a strong graded association with mortality, both good (RR 1.77; 95%CI 1.29–2.42), fair (RR 3.27; 95%CI 2.39–4.48) and poor (RR 7.52; 95%CI 5.20–10.88) self-rated health indicating a higher probability of death than

excellent self-rated health. After controlling for the characteristics of congestive heart failure, blood pressure, use of diuretics, posterior tibial artery blood pressure, albumin, creatinine, forced vital capacity, abnormal ejection fraction, aortic stenosis, major ECG abnormality, maximum stenosis of internal carotid artery, digit symbol test, instrumental activities of daily living, socio-economic characteristics and life-style factors (all significant independent predictors in the Cox proportional hazards model), mortality was elevated only for those with poor self-rated health (RR 1.91; 95%CI 1.27–2.87) and there was no difference between the other categories. The study was not focused on self-rated health and the authors did not discuss this finding. The findings were very similar in a five-year follow-up in the Established Populations for Epidemiologic Studies (EPESE) cohort, which also included a wide array of objectively confirmed clinical and laboratory data (Jylhä et al., 2006). These studies, together with others with a smaller range of health variables, indicate that the predictive power of self-rated health is dramatically weakened when clinically verified diseases and other objectively measured health indicators are controlled. On the other hand, psychosocial variables such as social support, neuroticism, locus of control or positive or negative affect do not seem to further contribute to the relationship of self-rated health with mortality (Benyamini et al., 1999; Idler & Kasl, 1991; Mackenbach, Simon, Looman, & Joung, 2002).

In the two studies described above (Fried et al., 1998; Jylhä et al., 2006), in spite of the wide array of reliable health indicators, self-rated health maintained an independent, albeit weakened statistical association with mortality: when all other characteristics were controlled for, those with “poor” self-rated health still showed an increased likelihood of dying during the follow-up period. In many studies researchers have been puzzled by similar findings. There is little evidence to support the widely discussed hypothesis (DeSalvo et al., 2006; Ferraro, Farmer, & Wybraniec, 1997; Idler & Benyamini, 1997) that the residual effect could be explained by “non-health” factors, such as the influence of self-rated health on future health behavior, or its ability to reflect other psychic and social resources or dispositional optimism or pessimism. Health behavior, social resources and an optimistic disposition (Giltay, Geleijnse, Zitman, Hoekstra, & Schouten, 2004) are known to be associated with mortality, but *through* their influences on the biological organism, *not independently* of them (Stephens, Wardle, & Marmot, 2005). Therefore, they would contribute to the residual effect of self-rated health if they reflected changes in objective health status that take place between the time of self-rating and death. This is a possible but not a very likely hypothesis, undermined by findings in many (Ben-Ezra & Shmotkin, 2006; Benyamini, Blumstein, et al., 2003; Murata, Kondo, Tamakoshi, Yatsuya, & Toyoshima, 2006; Nielsen et al., 2008; Singh-Manoux et al., 2007) but not all (Deeg & Kriegsman, 2003; Vuorisalmi, Lintonen, & Jylhä, 2005) time comparisons that the association of self-rated health is stronger for shorter than longer follow-up periods.

The most plausible explanation for the association between self-rated health and mortality that persists after controlling for all available health information seems to be that “it is a very inclusive measure of health reflecting health aspects relevant to survival which are not covered by other health indicators” (Mackenbach et al., 2002, p. 1162; Idler & Benyamini, 1997). Basically, the reason why all relevant aspects usually are “not covered” lies either in the practical limitations of empirical studies, or in the inadequacy of our present knowledge to appropriately measure these aspects. The variety of health variables that can be included even in the best studies is always limited, and can never exhaustively cover the variety of health problems among participants in a large population study. The measures of disability and functional status can rarely capture the whole range of functioning in a study sample, which is likely to be taken into account in self-ratings. The selection of

disease diagnoses inquired cannot be too large, and many diseases experienced by the participants will necessarily be left out. In self-ratings, the individual is likely to take into account the severity, consequences and the expected prognosis of the disease; this information is hardly ever collected in population studies.

One potentially vitally important but little studied pathway from physiological states to self-rated health consists of bodily sensations. Researchers in neurobiology, neuroendocrinology, immunology and cognitive neuroscience are increasingly interested in afferent information that conveys messages from the organism to the brain (Cameron, 2001; Craig, 2003; Danzer, 2001). This information does not necessarily reach consciousness; in fact, most of it participates in the regulation and adaptation of physiological functions at lower levels of the central nervous system. To some extent, however, these biological messages are perceived by the individual as sensations, feelings and emotions. The concept of ‘interoception’ is often used to refer to this sense which reflects the physiological condition of the entire body, the “afferent information that arises from anywhere and everywhere within the body – the skin and all that is underneath the skin” (Cameron, 2001, p. 697; Craig, 2003). The sense of position and movement (proprioception) and the sensations from internal organs (visceroception) are relatively well-known. According to Cameron (2001, p. 704) the most extensively studied interoceptive process is awareness of cardiac action, both heart rate and individual heart beats.

As far as self-rated health is concerned, the most interesting new line of research deals with conscious representations of humoral, biochemical bodily processes, and particularly those belonging to the innate immunological defense response. It is now known that a family of proteins called cytokines is centrally involved in different inflammatory processes, which in turn have a major role not only in infectious conditions but also in the pathogenesis of common chronic diseases such as cardiovascular disease, Alzheimer’s disease, diabetes and cancer. The level of many cytokines is elevated with normal biological aging. Increasing evidence shows that inflammatory processes and certain cytokines are associated with symptoms such as tiredness, impaired sleep, depressive mood, general malaise and poor appetite (Dantzer, O’Connor, Freund, Johnson, & Kelley, 2008). These symptoms are common in acute infections but may also appear without any diagnosed disease, reflecting sub-clinical dysregulation of the organism. For instance, chronic unexplained fatigue is now largely understood as a result of immune dysregulation (Klimas & Koneru, 2007).

Research on interoception and the biological basis of several frequent symptoms justifies the hypothesis that interoceptive processes constitute one pathway through which information of bodily states is conveyed and included into self-rated health. This is not a new hypothesis. As early as 1964, Stenback suggested that information received through interoceptive sensations may contribute to self-rated health. Kaplan and Camacho (1983) hypothesized that psychoendocrinological and psychoneuroimmunological processes play a part in the mediation between self-rated health and mortality, and subsequently several others have considered the possibility that different bodily feelings that may even be difficult to verbalize may contribute to self-assessments and thus to the association of self-rated health and mortality (Idler & Kasl, 1991; Uden et al., 2007).

The empirical evidence is still weak and haphazard, however. It is known that symptoms such as chronic pain (Mantyselka, Turunen, Ahonen, & Kumpusalo, 2003) and fatigue (Molarius & Janson, 2002) are important components of self-rated health. Recent studies have found associations between self-rated health and several inflammatory markers measured from blood, particularly cytokines such as interleukin-1 β , tumor necrosis factor alpha (TNF- α) (Lekander, Elofsson, Neve, Hansson, & Uden, 2004; Uden et al.,

2007) and interleukin-6 (Janszky, Lekander, Blom, Georgiades, & Ahnve, 2005). In one study, a graded association was observed between lower hemoglobin, higher white cell count, lower albumin and poor self-rated health even for values that are not considered pathological (Jylhä et al., 2006). In the analyses, several clinically confirmed diseases and other indicators were adjusted for. The respondents were not likely to be aware of the measured values of these markers or their clinical interpretation, and therefore could not have used them in their assessments. It is possible that the markers reflect the unmeasured severity of different health problems of which the participants were aware. However, an alternative hypothesis refers to the direct interoceptive route, suggesting that even bodily irregularities and dysfunctions too minor to be defined as medical conditions or conceptualized as illness can be sensed, are interpreted as indicators of health status, and are incorporated into self-assessments. With further empirical evidence, this pathway could significantly add to our understanding of the specific features of self-rated health and its predictive power.

To conclude, there are good reasons to believe that self-rated health is an “independent” statistical predictor of mortality because it is a more inclusive measure than other direct health indicators used in population studies. I suggest that the advantage of global self-ratings comes partly from the fact that empirical studies are necessarily constrained in the range and number of health variables they can include, and partly from the individual's superior ability to employ, more or less consciously, subtle and even non-conceptualized sensations of his or her bodily status in self-ratings of health. The latter is still more of a hypothesis than a conclusion based on firm empirical evidence.

Use of the self-rated health variable in research and clinical practice

It is time now to ask what can be said about the use of self-rated health in research, clinical practice and health policy, and about the validity of the measure. What did the early pioneers mean when they said that “...depending upon one's hypothesis such a self-rating may or may not be valid” (Suchman et al., 1958, p. 232)?

In his discussion about the validity of health assessments, Hyland (1993) emphasizes that validity should not be understood as an abstract term but established with reference to particular use, and that a measure can be valid in more than one way. As I argued earlier in this paper, the absence of a direct objective measure of “true health” means that there is no gold standard, no clear criterion for the validity of self-rated health. Even mortality is not a perfect criterion if one is not prepared to reduce health solely to the probability of survival (Lee et al., 2007). Here, the question is approached from a practical perspective: where and how should self-rated health be used?

Self-rated health has been used and recommended for use in comparisons of health status between population groups; as an outcome variable in clinical trials; and as a routine indicator in risk assessments and clinical practice. For all these purposes, the main advantages and most important limitations of its use arise from the unique nature of the measure: it is comprehensive, it is inclusive, and it is absolutely non-specific. These features allow it to capture dimensions of health that could not be captured by more detailed or guided questions, but at the same time the researcher has no control over which aspects of health are emphasized in the individual assessment.

In population studies, self-rated health is probably the most feasible, most inclusive and most informative measure of health status. In a given cultural environment, it is a powerful predictor of future health and use of health services. However, distributions of self-rated health are not directly comparable between cultural

groups and distant age groups; the same levels of self-rated health may imply different levels of “objective” health status in different groups, even if the associational structures of self-rated health were generally similar (Bardage et al., 2005; Jylhä et al., 1998). Using the SHARE data, Juerges (2007) tried to separate the influence of “true health” from the influence of response styles in between-country differences by using the other health measures available in the study to “correct” the self-ratings. The problem with this interesting exercise is that it ignores the variance in health not measured by the “objective” indicators and the complexity of the evaluation frames; it is not clear that the corrected measures are any more comparable than the original ones. By now, it seems clear that cultural and country comparisons of self-rated health should be based not on assumptions of identical correspondence between self-ratings and objective health indicators, but rather on the assumption of their context-specific relevance and validity.

Researchers should also be aware of the different meanings of crude, unadjusted values of self-rated health and values that are adjusted for other available health indicators. The contents of the adjusted values depend on the variables that were controlled for; differences between two population groups in self-rated health that is adjusted for a comprehensive set of medical and health variables may reflect group-specific evaluation frameworks and response styles rather than health status. The differences between crude and adjusted measures are most striking in comparisons between age groups where the non-adjusted values of global self-rated health usually indicate worsening health, while adjusted values indicate improving health with increasing age.

Distributions of global and comparative questions are not directly comparable. In older populations, one study found that the comparative question (Manderbacka, Kåreholt, Martikainen, & Lundberg, 2003) and another study that the global question (Vuorisalmi et al., 2006) was a better predictor of mortality. However, the comparative question is likely to be more influenced by the age-dependent frames of evaluation (Vuorisalmi et al., 2005), meaning that in relation to objective health indicators, comparability between age groups is even worse than with the global question. Therefore, at least in studies with a broad age range, preference should be given to the global question.

In clinical trials and clinical practice, self-rated health provides a measure that is genuinely sensitive to the patient's perspective and can significantly complement more specific measures (Bjorner, Fay-ers, & Idler, 2005). In patients with advanced cancer, self-rated health is a stronger predictor of mortality than performance and selected clinical indicators, symptoms, and health-related quality of life measures (Shadbolt, Barresi, & Craft, 2002). It detects different clinical states (Crane et al., 2006) and predicts functional decline (Fleishman & Crystal, 1998) and survival in HIV patients (Dzekedzeke, Siziya, & Fylkenes, 2008). It also shows a treatment effect after coronary angioplasty (Pocock et al., 2000). By now there is enough evidence to recommend self-rated health as one complementary outcome measure in clinical trials and clinical practice, alongside other, more specific measures. Additional information is received if, in addition to general self-rated health, patients are directly asked to assess how their health status has changed during a certain period of time, or to predict its development in future (Wang & Satariano, 2007).

In general practice and risk assessment, self-rated health is a useful and convenient tool. It adds predictive value to traditional risk factors such as diabetes, smoking, BMI, high cholesterol and hypertension in predicting stroke (Emmelin et al., 2003), and to the Framingham tool of coronary heart disease and cardiovascular death (May et al., 2006). In general practice, the self-rated health question can serve as a comprehensive “screening” tool for patients' health status. Also, it is likely to support the doctor–patient relationship in allowing the patient to generate

a perspective genuinely his or her own. Because of the different contextual frameworks of evaluation, positive self-rated health is no guarantee of good physical health, but poor self-rated health certainly warrants further attention. In the evaluation of treatment effects and in general practice, self-rated health can hardly replace more specific assessment tools, but it does yield information that cannot be reached by any specific tool.

Next step towards a better understanding of self-rated health

This paper suggests that (1) self-ratings of health are produced in a cognitive process that is inherently subjective and contextual, and (2) the basis of self-rated health lies in the biological and physiological state of the individual organism, and this explains its association with mortality. The paper has also proposed a conceptual model that can help understand how these two seemingly contradictory claims could both be true. In their seminal paper, Idler and Benyamini (1997) argued that researchers are approaching the limits of what secondary analyses of large longitudinal data sets, often planned for other purposes, can tell us about the relationship between self-rated health and mortality. They suggested that the next steps of research should include (1) outcomes other than mortality, (2) special populations, (3) qualitative approaches and (4) the cognitive/cultural processes associated with assessments of health. Twelve years on, our understanding in all four directions has improved, but most findings still come from studies that have not been primarily designed to examine self-rated health.

Two major challenges can be identified for future research: the first is to gain a better understanding of the cognitive processes of health assessments; and the second to gain a better understanding of the role of the unique information people have direct from their own body. To some extent it is possible to use secondary analyses in available large data sets, but it is also necessary to conduct studies specially planned for this purpose.

1. Study the ways in which people reason about their health and the logics that govern their consideration of different components of health in relation to self-ratings

Much is already known about the aspects of health and well-being that people take into account in their self-ratings and about the relative importance of these aspects in different population groups and stages of life. Still, it is important to know not only *which* factors are taken into account in self-ratings, but also *how* they are taken into account. Therefore, it would be worthwhile to use qualitative approaches to analyze people's own explanations and accounts of the process of reasoning that leads them from reviewing different aspects of their health to the conclusion that in general, their health is "good" or "poor". Focusing on the different logics of reasoning could help us understand the differences in the evaluation frames between population groups.

2. Study the individual bodily sensations, their interpretations and their possible biological background

The role of direct bodily sensations, conceptualized as symptoms, ailments, feelings, etc., has received much less research attention than the role of diseases and functioning, and it is poorly understood. One of the hypotheses put forward in this paper is that, through these sensations, self-rated health can capture subtle bodily information that is not necessarily represented as diagnosed health conditions, and that this may contribute to the association of self-rated health with mortality. In empirical research, this hypothesis can be approached from two directions:

- a) Study the experience and role of bodily sensations in self-rated health

The nature of these sensations may be diffuse and variable. They constitute an important bridge between the state of the organism and the consciousness of an individual, and are likely to play a central role in conveying information about different regulative mechanisms into consciousness. Qualitative analyses, inviting people to describe and verbalize their bodily sensations and interpret them in relation to their health status, would help create a new body of information on factors underlying self-rated health.

- b) Study the role of objectively measured biological and physiological parameters in self-rated health

To better understand the biological basis of self-rated health, we need to know more about its associations with different biomarkers, particularly those involved in inflammatory, psychoendocrinological and psychoneuroimmunological processes. For this purpose, secondary analyses of existing population studies would be appropriate.

To conclude, self-rated health is a unique, valuable indicator of human health status. Its origins lie in a process where information from the individual's body and mind is received, selected, reviewed and summarized. If we are to gain a closer understanding of this process, it is necessary for biological and social sciences to pool their resources and work more closely together.

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